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Original Paper

Data and text mining

Discovering drug–target interaction knowledge from biomedical literature

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Abstract

Motivation: The interaction between drugs and targets (DTI) in human body plays a crucial role in biomedical science and applications. As millions of papers come out every year in the biomedical domain, automatically discovering DTI knowledge from biomedical literature, which are usually triplets about drugs, targets and their interaction, becomes an urgent demand in the industry. Existing methods of discovering biological knowledge are mainly extractive approaches that often require detailed annotations (e.g. all mentions of biological entities, relations between every two entity mentions, etc.). However, it is difficult and costly to obtain sufficient annotations due to the requirement of expert knowledge from biomedical domains.

Results: To overcome these difficulties, we explore an end-to-end solution for this task by using generative approaches. We regard the DTI triplets as a sequence and use a Transformer-based model to directly generate them without using the detailed annotations of entities and relations. Further, we propose a semi-supervised method, which leverages the aforementioned end-to-end model to filter unlabeled literature and label them. Experimental results show that our method significantly outperforms extractive baselines on DTI discovery. We also create a dataset, KD-DTI, to advance this task and release it to the community.

Availability and implementation: Our code and data are available at https://github.com/bert-nmt/BERT-DTI.

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Supplementary information: Supplementary data are available at *Bioinformatics* online.

1 Introduction

Discovering knowledge from biomedical literature is an important task for both research organizations and industrial companies. PubMED, one of the most famous search engines for biomedical literature (https://pubmed.ncbi.nlm.nih.gov), has indexed more than 30M articles, and millions of new papers came out every year (Landhuis, 2016). It is impossible to manually check all the papers to obtain useful knowledge, which results in an urgent demand to automatically discover knowledge from the literature.

The interaction between drugs and targets (DTI) in the human body plays a crucial role in biomedical science and applications (Sachdev and Gupta, 2019; Wu *et al.*, 2018; Yamanishi *et al.*, 2008), e.g. drug discovery, drug repurposing, precision medicine, etc. In biomedical literature, a drug refers to any type of medication, ranging from small molecules like Aspirin, Penicillin to large molecules like Hepatitis B Vaccine. A target could be protein, enzyme or nucleic acid in our body, which binds the drugs we take. Drugs interact with targets in different ways. For example, Aspirin (drug) can inhibit (interaction) COX-1 (target), and Streptokinase (drug) can activate (interaction) Plasminogen (target). For simplicity, we call the triplet of Drug, Target and their Interaction as a 'DTI triplet'.

Discovering DTI triplets from biomedical papers is challenging. First, lots of terms and aliases (e.g. abbreviations and synonyms) exist in an article, but only a small set of them contributes to DTI triplets, which makes this task harder than conventional relation extraction from general text. As shown in Figure 1, given the title and abstract of a paper, we want to discover the triplet ⟨Clotrimazo, Ergostero, Inhibitor⟩. We can see that there are many terms like 'fungal cytoplasmic membrane', 'azoles' and 'C. albicans', which are not related to the triplet we want to discover and increase the difficulty of the task. Second, lots of DTI knowledge is expressed by multiple sentences (Verga *et al.*, 2018), which is more difficult compared to the conventional relation extraction built upon one or two sentences (Yao *et al.*, 2019). As shown in Figure 1, no single sentence contains a complete DTI triplet.

Existing methods of discovering biological knowledge are mainly extractive approaches that usually sequentially use named entity

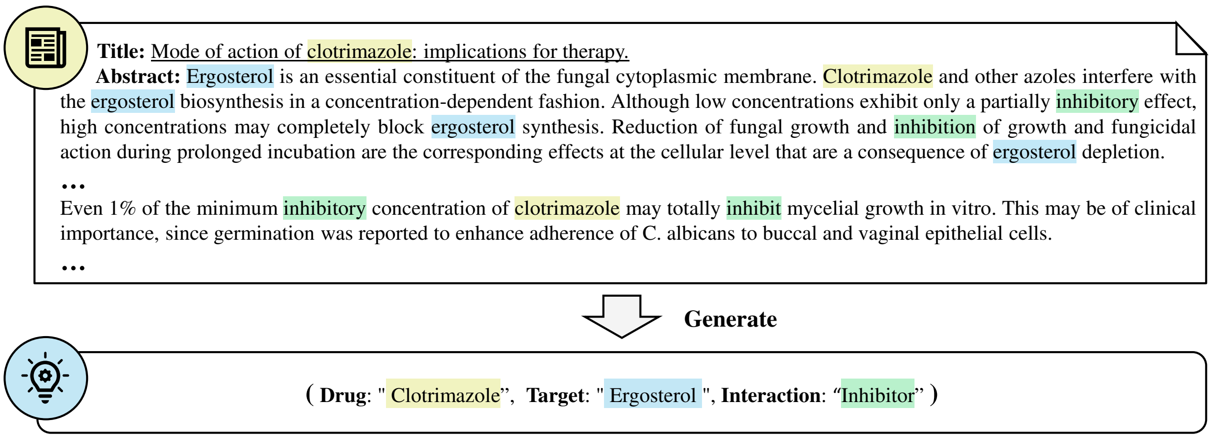

Fig. 1. Examples of drug–target–interaction knowledge discovery from literature. This abstract is from Haller et al., (1985).

recognition and entity relation extraction. Learning models for such pipeline requires detailed annotations, including all mentions of biological entities, relations between every two entity mentions and so on. However, it is difficult and costly to obtain sufficient annotations. On one hand, since the literature is usually long, the labeling workload is greatly increased compared to common knowledge extraction whose input is relatively short (e.g. a few sentences). On the other, due to the diverse term-entity aliases and specialized expressions, identifying DTI triplets from papers requires expert knowledge of biomedical domains, as shown in Figure 1.

To rescue this task from the annotation difficulties, in this article, we explore an end-to-end solution. We regard the DTI triplet as a sequence following the order of drug, interaction and target, and use a Transformer-based model for generation. The document is fed into the encoder, and the DTI triplet is directly generated at the decoder. Therefore, learning such model does not need to annotate every entity mentions and their relations. The PubMedBERT (Gu *et al.*, 2020), a model pre-trained on 30M abstracts from biomedical literature, is applied to the encoder and decoder of our model. Further, we use semi-supervised annotations to improve performance, where the end-to-end model trained on the limited labeled data is used to select/filter the literature and label them.

Experimental results demonstrate that (i) generative models perform better than extractive ones and are more promising for this task; (ii) leveraging unlabeled data can further boost the performance of generative models; and (iii) the performance of all the models is far from industrial demands, even boosted by semi-supervised learning, which suggests that DTI discovery is a challenging task and calls for more research efforts from the machine learning and natural language processing community.

Finally, we provide a new dataset, KD-DTI, for discovering DTI from documents. KD-DTI is built upon DrugBank (Wishart *et al.*, 2018) and Therapeutic Target Database (TTD) (Wang *et al.*, 2020). As far as we know, KD-DTI is the largest dataset for discovering DTI triplet from literature. The dataset contains 12K training samples, 1K validation samples and 1.1*K* test samples. We will release the data to the community.

Our contributions are summarized as follows:

1. We explore an end-to-end generative solution for extracting DTI triplet from biomedical literature, which shows the possibility of discovering DTI knowledge with much less annotation effort (Section 2). The method can be further enhanced by semi-supervised learning as we proposed (Section 2.2).
2. We create KD-DTI (Section 3.1), the largest dataset for discovering Drug–Target–Interaction triplets from literature. It contains 14K labeled data and 3671K words in total. We expect that such a dataset will advance the research of knowledge discovery from biomedical literature.
3. We study the performance of current methods (Section 4) and point out future directions for the DTI discovery task (Section 6).

2 Our method

In this section, we first introduce our end-to-end generative solution for DTI triplet discovery (Section 2.1), and then describe how we utilize unlabeled data to further ease the difficulty of annotation (Section 2.2).

2.1 End-to-end triplet generation

To avoid labeling intermediate annotations (i.e. labels for entities mentions and relations between each pair of entities) and sequentially applying multiple models as extractive methods, we explore generative methods for this task. Specifically, we use a Transformer model (Vaswani *et al.*, 2017). The encoder of Transformer is used to encode the document (i.e. title and abstract), and the decoder of Transformer works for generating the DTI triplets. The output of the decoder follows the following format:

$$< \mathtt{d} > \text{drug}_1 < \mathtt{i} > \text{interaction}_1 < \mathtt{t} > \text{target}_1$$
$$< \mathtt{d} > \text{drug}_2 < \mathtt{i} > \text{interaction}_2 < \mathtt{t} > \text{target}_2 \cdots,$$

where the drug, interaction and target are separated with special tokens $< \mathtt{d} >$, $< \mathtt{i} >$ and $< \mathtt{t} >$, and all triplets are concatenated as a longer sequence.

Recently, pre-trained language models, such as BERT (Devlin *et al.*, 2019) for common domain and PubMedBERT (Gu *et al.*, 2020) for biomedical domain, achieve great success in NLP areas. However, as suggested by previous work (Dong *et al.*, 2019; Zhu *et al.*, 2020), directly using BERT to initialize parameters of generation models is not the best choice {Using generative pre-trained models [e.g. GPT2 (Radford *et al.*, 2019)] may avoid such effects, and we leave it as future work.}. Therefore, we follow Zhu *et al.* (2020) to indirectly incorporate the pre-training models.

For encoder, we first use BERT to extract token representations as $B = \{\boldsymbol{b}_1, \boldsymbol{b}_2, \ldots, \boldsymbol{b}_n\}$, where $\boldsymbol{b}_i$ is BERT embedding of the ith token. After that, given an L-layer Transformer encoder, we obtain final encoder output by incorporating additional attention over the features extracted by BERT. Mathematically, the outputs of the lth encoder layer, H_l, is calculated as follows:

$$\begin{aligned} A_l &= \frac{1}{2}(\text{Attn}(H_{l-1}, H_{l-1}) + \text{Attn}(H_{l-1}, B)), \\ \tilde{H}_l &= \text{LN}(H_{l-1} + A_l), \\ H_l &= \text{LN}(\tilde{H}_l + \text{FFN}(\tilde{H}_l)), \end{aligned} \tag{1}$$

where $H_0 = \{\boldsymbol{e}_1, \boldsymbol{e}_2, \ldots, \boldsymbol{e}_n\}$ is the embedding of all input tokens [H_0 in Equation (1) and H'_0 in Equation (2) are not the BERT embedding

B, but the input embedding of the Transformer encoder/decoder, which are randomly initialized and learned during training.]. LN is the layer normalization. $\mathrm{Attn}(\boldsymbol{Q}, \boldsymbol{V})$ is the multi-head self-attention function operating on vector packages of queries $\boldsymbol{Q}$, values $\boldsymbol{V}$ (also used as keys). FFN is a position-wise feed-forward network. We denote $R = H_L$ as the final representations of input tokens.

On the decoder side, we use the attention models over BERT features in a similar way to the encoder. The hidden state of j-th decoding step is calculated as:

$$\begin{aligned}
\tilde{A}'_l &= \mathrm{LN}(H'_{l-1} + \mathrm{Attn}(H'_{l-1}, H'_{l-1})), \\
A'_l &= \frac{1}{2}(\mathrm{Attn}(\tilde{A}'_l, R) + \mathrm{Attn}(\tilde{A}'_l, B)), \\
\tilde{H}'_l &= \mathrm{LN}(\tilde{A}'_l + A'_l), \\
H'_l &= \mathrm{LN}(\tilde{H}'_l + \mathrm{FFN}(\tilde{H}'_l)),
\end{aligned} \tag{2}$$

where H'_l is the output of the l-th decoding layer, which is a package of j hidden state representations. $H'_0 = \{e'_1, e'_2, \ldots, e'_j\}$ is the non-BERT embedding of all decoded tokens plus a <SOS> token (input of the first decoding step).

2.2 Semi-supervised learning

The end-to-end solution proposed in the previous section reduces the efforts of detailed annotation, like annotating where an entity starts/ends in a paper (even it does not contribute to the final DTI), or the evidence that how a relation is obtained. However, it is still cost to obtain a large amount of the (document, DTI triplet) pair, since such labeling requires solid biomedical background. To remedy this, we proposed a semi-supervised method based on our end-to-end model. The method consists of two steps, where we first filter the data based on rules, and then use the model in the previous section to refine the labels. We also discuss the relation with distance supervision (DS).

(Step-1) *Rule-based filtration*: We download all available titles and abstracts from PubMED. For each document (i.e. title and abstract), we use ScispaCy (Neumann *et al.*, 2019), an open-sourced NER tool to find out all possible drug and target entities, and use `FuzzyMatch` to search all mentioned interactions in the document (We collect commonly used interactions from DrugBank in advance, like inhibition, agonist, antagonist, etc.). Assume, we extract m, n and p drugs, targets and interactions from document A. By enumerating all permutations, document A corresponds to $m \times n \times p$ $\langle$drug, target, interaction$\rangle$ triplets. An underlying assumption is that the DTI triplets that appear more frequently across all documents have better confidence to be true. We then count the numbers of occurrences of each DTI triplet and delete DTI triplets with <10 occurrences (According to our preliminary exploration, if we randomly select a drug, a target and an interaction from our dataset, most of those DTI triplets occur <4 times in all the PubMed papers.). We keep the documents that have at least one DTI triplet after the deletion. We eventually obtain a dataset with 139K documents, which is denoted as $\mathcal{D}_{\mathrm{semi}}$.

(Step-2) *Model-based labeling*: We propose a method based on knowledge distillation (Hinton *et al.*, 2015) to refine the DTI triplet in $\mathcal{D}_{\mathrm{semi}}$. First, we use a generative model trained on supervised data to generate DTI triplets for each unlabeled document. If the Transformer model does not generate any triplet for a document from $\mathcal{D}_{\mathrm{semi}}$, we remove the document from $\mathcal{D}_{\mathrm{semi}}$. Each remaining document in $\mathcal{D}_{\mathrm{semi}}$ is associated with at least one generated triplet and at least one pseudo triplet. If at least two elements (e.g. drug–target, drug–interaction or target–interaction) of a pseudo triplet are the same as those of a generated triplet, we keep this document (and the matched triplet in $\mathcal{D}_{\mathrm{semi}}$); otherwise, we delete it. [We choose to match two rather than all three, because: (i) we follow the famous setting of distance-supervision (Mintz *et al.*, 2009), where two elements of triplets are matched. (ii) There are too few samples left when matching all three elements. (iii) `FuzzyMatch` cannot cover all element mentions, and therefore, matching two can leave some leeway and improve the recall of the missing cases.] After filtration, there are $5.8K$ documents left in the dataset. Denote this dataset as $\mathcal{D}_{\mathrm{KD}}$. Note we will use the matched pseudo triplets in $\mathcal{D}_{\mathrm{semi}}$ for the following experiments; the generated triplets are only used for filtration, but not for model training.

Another way to use unlabeled data is DS (Mintz *et al.*, 2009). Given any DTI triplet (d, t, i) from the labeled dataset, we use `FuzzyMatch` to search all $\overline{D}_j$ in $\mathcal{D}_{\mathrm{semi}}$. If we find mentions of both d and t, we assign a pseudo label/triplet (d, t, i) to $\overline{D}_j$. Denote the obtained dataset as $\mathcal{D}_{\mathrm{DS}}$, which has 15K samples. We empirically found that our proposed semi-supervised method is better than DS, since DS does not take interactions into consideration.

3 The corpus

In this section, we first introduce the acquisition of the proposed dataset KD-DTI (Section 3.1), and then introduce its statistics and characteristics (Section 3.2). The dataset is available in the Supplementary Material and will be released later.

3.1 Dataset creation

Data collection. The DTI triplets in our dataset come from two widely used databases, DrugBank (Wishart *et al.*, 2018) and TTD (Wang *et al.*, 2020). (i) DrugBank is a pharmaceutical knowledge base that consists of proprietary authored content describing clinical-level information about drugs (https://go.drugbank.com/). DrugBank covers 14 315 drugs, 4885 targets, 63 types of interactions and 18 866 DTI triplets. (ii) TTD (http://db.idrblab.net/ttd/) is a comprehensive collection of various types of drugs, which includes 37 316 drugs, 3419 targets, 109 interactions and 43 874 DTI triplets. Given a DTI triplet, if the reference papers are provided and the abstracts of those papers are openly accessible, we record the triplet and the paper. As the first step, we only use the titles and abstracts of the reference papers. The dataset of this step is denoted as $\mathcal{D}$.

Data filtration. As a starting point of structured DTI knowledge discovery, we are only interested in the document, which contains enough information to discover a DTI triplet. However, in $\mathcal{D}$, some papers only generally describe some drugs and targets, in which the DTI triplets do not explicitly appear. Therefore, we heuristically filter out the samples in $\mathcal{D}$ by which we cannot obtain the associated DTI triplets. The basic idea is that we require that the drug, target and interaction in a triplet should be all included in a paper (name or alias appear). After filtration, we obtain $1.3K$ documents as test set, 1K documents as the validation set and 12K documents as the training set. The detailed filtration process is described in Supplementary Appendix B.

Human verification. We then manually check all the samples in the test sets. We employ eleven annotators with Ph.D. background. Each (document and DTI triplets) pair is independently checked by two annotators. If their evaluation results are different, another two annotators are involved for discussions. We remove those difficult cases that a consensus is not reached after the discussions of four annotators. We eventually obtain 654 (document and DTI triplets) pairs from DrugBank, and 505 pairs from TTD for the test set. We explicitly split them into two parts so that one could check the performance on different data sources.

3.2 Dataset statistic

Table 1 shows the statistics of our dataset as well as some related datasets. We have the following observations:

1. In terms of data size (including numbers of documents, sentences and words), our dataset is much larger than previous datasets.
2. Although ChemProt and CPI-DS also focus on tasks in the biomedical domain, they do not directly serve for drug–target interaction discovery from literature. ChemProt mainly focuses on relation extraction and assumes that entities are given in advance, while our KD-DTI is to discover DTI triplets (instead of relation only) from documents. CPI-DS is to extract relation

Table 1. Statistic of datasets for discovering biomedical knowledge

Dataset	No. of documents	No. of relations	No. of sentences	No. of words (K)	Knowledge
CPI-DS (Döring *et al.*, 2020)	N/A	1	2613	486	Chemical proteins relation
BC5CDR (Li *et al.*, 2016)	1500	1	11 089	282	Chemical disease relation
ChemProt (Antunes and Matos, 2019)	2432	5	24 923	650	Chemical proteins relation
KD-DTI	14 256	66	139 810	3671	Drug–target interaction
KD-DTI (semi)	139 408	66	1 556 614	39 997	Drug–target interaction

Note: 'KD-DTI (semi)' denotes semi-supervised dataset in Section 2.2.

from single sentences, and thus is much easier than our task that tackles long documents.

3. KD-DTI includes a rich variety of relationship types that are not covered by previous datasets. ChemProt covers five relations, and CPI-DS and BC5CDR contain one relation only. In comparison, there are 66 relations in our dataset.
4. KD-DTI is collected from more than one data source, i.e. DrugBank and TTD, which could be used to evaluate the generalization or transfer abilities of machine learning algorithms/models.

Besides, in the KD-DTI dataset, the average number of triplets for each document is 2.17, and there are 24K unique triplets. After adding the semi-supervised data, the number of unique triplets grows to 36K. The overlap ratio of target is DTI triplet is 28.8% for test set and 31.5% for valid set. The drug overlap is 42.5% for test set and 38.2% for valid set. And the target overlap is 73.2% for test set and 89.0% for valid set.

4 Experiments

In this section, we introduce several extractive baselines in Section 4.1, describe the settings of experiments in Section 4.2 and show the results of our end-to-end method and the results of using semi-supervised learning in Sections 4.3 and 4.4.

4.1 Extractive baselines

We compare with two extractive baselines: Cascade Relation extraction (CasRel), which is the state-of-the-art extractive method (Wei *et al.*, 2020) and a pure NER method, which regards relation as a special entity.

CasRel: CasRel is a cascade tagging method that can jointly perform NER and RE. CasRel leverages BERT to extract representations for input sequences. To find out DTI triplets, CasRel first tags out all possible drugs (i.e. subject) of the input. After that, CasRel searches interactions (i.e. relation) and targets (i.e. objective) for the discovered drugs. For this purpose, we train a classifier for each relation, whose input is the discovered drug and the output is the position of the target, i.e. classifications of whether each token is the start or end token for the target phrase. The classifier is allowed to output null, indicating no target for this relation.

To use CasRel, we obtain the named entity annotations of drugs and targets by searching the document with `FuzzyMatch`. Although CasRel achieved great success in standard relation-extraction tasks like NYT (Riedel *et al.*, 2010) and WebNLG (Gardent *et al.*, 2017), in our setting, the annotations for all entities are automatically obtained without manually checking, which limits the performance of CasRel.

Pure NER method: In biomedical literature, interactions often explicitly appear in documents with specific forms (e.g. noun, verb and past/present participle). Therefore, it is natural to regard the interaction as a special entity, and use a NER model to figure out the DTI triplets. For this purpose, after obtaining the mentions of drugs, targets and interactions using `FuzzyMatch`, we train a BERT-based NER model where interactions are also types of entity. During training, the BERT-based NER model is trained to predict the possibility of whether a token belongs to entity spans of drugs, targets and interactions. At the inference phase, the trained model tags out token spans of drugs, targets and interactions. For simplification, we choose the drug, target and interaction with the maximum probability to constitute the DTI triplet. With this method, we can predict at most one DTI triplet for each document.

4.2 Settings

Hyperparameters. For CasRel, we mainly follow the hyperparameters suggested by Wei *et al.* (2020). A modification to CasRel is that since our input text can be longer than 512 (after BPE, there are 762 abstract longer than 512 tokens, and 19 abstract longer than 1K tokens), we cut the document into several pieces, each with a length of 512. We use BERT to encode each piece and concatenate all the representations for further processing. We use 66 relation-classifiers in total, where each classifier is a single-layer feed-forward network with ReLU activation that taking BERT embedding as input. The drug and target identifier is a single-layer feed-forward network.

For the pure NER model, an entity classifier of the single-layer feed-forward network is applied after the BERT module. We jointly finetune the BERT and the classification heads. We use Adam optimizer (Kingma and Ba, 2015) with learning rate 10^{-5}. The mini-batch size is two sequences. The models are trained for 100 epochs with early stopping, where the training stops if the validation performance does not increase for 5 epochs.

For generative models, after tokenization, we apply BPE (Sennrich *et al.*, 2015) to both the source sequences and target sequences to reduce vocabularies. We set the number of layers as two, and the embedding dimension as 256. We use Adam optimizer with the `inverse_sqrt` scheduler. The learning rate is 5×10^{-4} and warm-up steps are 8K. The dropout and attention dropout of Transformer are set as 0.2 and 0.1. The label smoothing is set as 0.2. The batch size is 12K tokens per GPU. For the Transformer with pre-trained models, we explore two methods as introduced in Section 2.1. We try the conventional $\text{BERT}_{\text{base}}$ model and $\text{PubMedBERT}_{\text{base}}$ model, in which $\text{PubMedBERT}_{\text{base}}$ is trained using abstracts of all PubMed papers. All models are trained on a single V100 GPU. For full model training (i.e. with pre-trained language models), it takes about 15 h to converge on a single V100 GPU.

Evaluation metrics. We define a set of metrics to evaluate the performance of a model for DTI discovery, covering different granularity: (i) *triplet-level metrics*, which are used in previous knowledge discovery work (Maimon and Rokach, 2005; Yao *et al.*, 2019) and assess the correctness of a discovered triplet as a whole; (ii) *ontology-level metrics*, which evaluate the correctness and completeness of all discovered DTI triplets from a paper corpus as a whole; and (iii) *entity-level metrics*, which evaluate the accuracy of discovered drugs, targets and interactions, respectively. We describe the detailed definition of metrics in Supplementary Appendix C.

4.3 Results of the end-to-end method

The test results of DrugBank and TTD are reported in Table 2. The standard deviation of results in Supplementary Appendix E and the case study in Supplementary Appendix F. We have the following observations:

Table 2. Results of the DTI triplet discovery on DrugBank and TTD

DrugBank	Triplet level			Ontology level			Entity level (Acc.)		
	*F*1	*P*	*R*	*F*1	*P*	*R*	Drug	Target	Interact
CasRel	15.42	13.74	17.57	18.62	20.74	16.89	27.12	23.14	31.32
Pure NER	18.20	19.11	17.37	17.25	19.40	15.54	60.72	35.25	79.26
Transformer	27.41	28.38	26.50	25.49	26.64	24.43	53.05	52.86	79.54
Transformer + BERT	30.32	31.46	29.26	29.50	31.64	27.64	53.00	55.27	79.47
Transformer + PubMedBERT	34.82	35.88	33.82	32.87	34.73	31.22	55.73	58.91	82.11
Transformer + BERT-attn	34.60	35.50	33.74	33.26	35.50	33.74	56.80	55.12	79.82
Transformer + PubMedBERT-attn	36.97	37.82	36.16	34.32	36.64	32.28	57.33	58.69	82.59
TTD	Triplet level			Ontology level			Entity level (Acc.)		
	*F*1	*P*	*R*	*F*1	*P*	R	Drug	Target	Interact
CasRel	5.74	4.87	7.00	6.05	9.30	4.49	18.77	18.51	18.06
Pure NER	6.66	6.77	6.55	6.32	9.23	4.81	33.82	16.93	68.40
Transformer	6.32	6.73	5.96	5.75	6.41	5.22	10.89	56.53	87.43
Transformer + BERT	7.63	7.87	7.41	7.36	8.44	6.53	14.44	51.98	87.72
Transformer + PubMedBERT	7.81	8.28	7.41	7.11	7.83	6.52	12.83	58.47	86.93
Transformer + BERT-attn	8.34	8.42	8.27	7.59	8.14	7.10	15.46	53.37	87.03
Transformer + PubMedBERT-attn	8.88	9.21	8.57	7.87	8.83	7.10	14.60	61.97	89.50

Note: 'CasRel' and 'Pure NER' are extractive methods leveraging BERT, and the remaining are generative ones. '-attn' denotes using pre-trained language models in the attention manner.

1. Generative methods obtain better results than the extractive method (i.e. CasREL) on KD-DTI, in terms of triplet-level metric and ontology-level metric. One reason is that our task lacks manual annotation of intermediate labels, such as the BIO representations of all entities and relations among any two entities. We obtain such intermediate labels with FuzzyMatch, which are usually of poor quality and therefore impair the performance of extractive methods. For the DTI triplet discovery task, such intermediate labels are often hard to obtain, and we should keep exploring how to improve performances without intermediate labels.
2. For extractive methods, the pure NER method outperforms CasRel on triplet-level metric and entity-level metric. Specifically, for entity-level drug accuracy, the pure NER method even achieves the best result. This shows when intermediate labels are lacking and the relations among entities are comprehensive, simplifying this problem (like extracting only one triplet for a document) is another choice. However, the pipeline form of the all extractive models and the fuzzy matches in data pre-processing may lead to cascading errors. In many cases, the entities and relationships may not appear directly, which increases the learning difficulty of the extractive model. By contrast, generative models predict matching answers directly in the decoding space and are less subject to direct interference from such data issues.
3. Using pre-trained models is helpful for our task. Taking DrugBank as an example, for triplet-level *F*1, after using conventional BERT to initialize the encoder, the metric can be improved from 27.41 to 30.32. After using PubMedBERT, which is a model pre-trained on all abstracts of PubMed, we achieve an even higher *F*1 score, 34.82. This demonstrates the effectiveness of pre-training, especially in-domain pre-training.
4. The manner of using pre-trained models also matters. Comparing with directly initializing the encoder with a pre-trained model, we find that indirectly incorporating the pre-training model can further boost the performance: BERT-attn and PubMedBERT-attn obtain more than 4 and 2 point improvement over BERT and PubMedBERT, respectively.
5. The scores on TTD are lower than DrugBank because TTD is a harder dataset. To verify this, we calculate the minimal distance between drugs and targets: Given a document D and a DTI triplet (d, t, i), let P_d and P_t denote two sets, which are positions of drugs and targets obtained by `FuzzyMatch` in D. The distance is defined as $\min_{p_d \in P_d, p_t \in P_t} |p_d - p_t|$. For DrugBank and TTD, the average minimal distances over all test samples are 34 and 51, which shows that identifying the DTI triplet from TTD requires understanding a longer document.
6. While using pre-trained models achieves the best results, we observe that it suffers from overfitting: The *F*1 score on the training set is 70.29 for PubMedBERT, which is much higher than those on the validation set (23.33) and test set (22.9, the average score of two test sets). We find that simply using larger dropout or label smoothing does not help, which suggests better regularization techniques are needed for this task. More details are in Supplementary Appendix D.
7. We note that in addition to the mask-style language model we use, there are also some recent generative pre-trained language models for biomedical domains, such as the SciFive (Phan *et al.*, 2021). These generative models can also be adapted to our end-to-end triplet generation framework. To verify this, we test SciFive-based model on DrugBank with the same setting of Transformer+PubMedBERT-attn, and achieve precision, recall and *F*1 of 30.43, 27.49 and 28.88, which outperform the extractive baselines but lag the proposed models. Same setting here denotes using the same amount of learnable parameters as that in our 'Transformer+PubMedBERT-attn', which is equivalent to two learnable layers in SciFive. When further increasing learnable parameters, SciFive-based model achieve 36.41 *F*1, which is comparable to our models.

Effect of generation order. Our methods learn to generate drug–target–interaction triplets sequentially for generative methods. An advantage of this method is that we could leverage the dependency among the triplets to improve the generation quality. A question arises: does the order of elements in DTI triplet matter? To find it out, we enumerate all six orders of the triplet on the standard Transformer model and the PubMedBERT-attn model. The results are in Table 3. Generally, the order of (drug, interaction and target) performs better, indicating that the order of triplet should be consistent with natural language order (i.e. subject–verb–object).

4.4 Results of the semi-supervised method

As generative models perform better than extractive ones, we focus on generative ones in this sub-section and conduct experiments with the Transformer model and Transformer + PubMedBERT-attn model. We merge the KD-DTI corpus with $\mathcal{D}_{\text{DS}}$ and $\mathcal{D}_{\text{KD}}$, respectively, to get two enlarged datasets, and then train models on them. Instead of training from scratch, we find that initializing the parameters from a model trained on the parallel corpus KD-DTI is better.

The results are shown in Table 4. We have the following observations:

1. Enhanced with our semi-supervised method ($\mathcal{D}_{\text{KD}}$), we achieve more than two points improvement on DrugBank, for both Transformer and Transformer + PubMedBERT; on TTD, significant improvements are also observed.
2. Enhanced with $\mathcal{D}_{\text{DS}}$, the generation performance is also generally improved, but not as much as $\mathcal{D}_{\text{KD}}$, which shows that the quality of the synthetic data is not as good as that from knowledge distillation. This is consistent with the discovery in Yao *et al.* (2019). Our conjecture is that the documents are rich in entities and noises, and simply using DS without a scoring mechanism cannot lead to significant improvement, especially when the model equips pre-trained knowledge.

   From observations (1) and (2), we can also conclude that pre-training and assigning pseudo labels to the unlabeled data are two orthogonal ways, both of which deserve more attention in the future.
3. We also directly combine $\mathcal{D}_{\text{semi}}$ with the parallel KD-DTI dataset (which is up sampled by five times) and get the largest training dataset in our experiments. However, while training Transformer (without BERT) with this large dataset, the triplet-level *F*1 scores on DrugBank and TTD are 18.19 and 3.15, respectively, which are much worse than training on KD-DTI only. This demonstrates the necessity of quality control in data enhancement.
4. Even if data enhancement can boost DTI discovery, the overall accuracy is still not very high. That is, DTI discovery is a challenging task. We need to design better models, algorithms and/or semi-supervised learning methods to meet the expectation of real-world applications.

Ablation analysis. To better understand the proposed semi-supervised method, we remove the *rule-based filtering* step and *model-based labeling* step, respectively. The result is shown in Table 5. We can see that when removing any of the proposed filtering steps, the performance drops significantly on both DrugBank and TTD data. These ablations show the effectiveness of two proposed filtering mechanisms and demonstrate a great necessity for reducing noise in semi-supervised data. We also compare with a variant of DS, where all of drug, target and interaction should appear in the input document (denoted as 'DS w/interaction' in Table 5). We can see such a variant of DS does not bring significant improvement compared to the standard DS.

4.5 Results of relation extraction

Our end-to-end generation framework can also be adapted to relation-extraction tasks. To achieve that, the input document is encoded by the encoder, and the two entities are used as the prefix of the decoder. The decoder will output the relation of the two entities.

We test the model performance on the ChemProt dataset (Antunes and Matos, 2019) and the results are presented in Table 6. Baseline results are taken from previous works (Lim and Kang, 2018). The results show that when no additional data pre-processing methods are used, our method significantly outperforms the dedicated document-level relation-extraction baseline. Note that our model is not designed for the relation-extraction task of ChemProt, where model is only asked to predict the relation given clearly extracted entities. Nevertheless, our method can still achieve better or comparable results to relation-extraction methods that use additional pre-processing.

Table 3. Results of the generation with different triplet orders

Triplet order		D-I-T	D-T-I	I-D-T	I-T-D	T-I-D	T-D-I
DrugBank	Transformer	**27.41**	26.34	26.66	25.15	25.38	25.18
	Transformer + PubMedBERT-attn	**36.97**	36.13	32.48	33.75	34.89	36.54
TTD	Transformer	**6.32**	5.01	5.87	4.72	5.81	4.37
	Transformer + PubMedBERT-attn	**8.88**	8.52	7.83	7.28	8.42	7.12

Bolded font indicates the highest score.

Table 4. Comparison of semi-supervised methods

DrugBank	Triplet level			Ontology level		
	No enhance	+ DS	+ Ours	No enhance	+ DS	+ Ours
Transformer	27.41	29.92	30.57	25.49	27.26	28.04
Transformer + PubMedBERT-attn	36.97	35.11	39.78	34.32	35.15	38.87
TTD	**Triplet level**			**Ontology level**		
	No enhance	+ DS	+ Ours	No enhance	+ DS	+ Ours
Transformer	6.32	6.99	7.20	5.75	6.19	6.66
Transformer + PubMedBERT-attn	8.88	10.83	12.26	7.87	8.01	9.80

Table 5. Analysis for the semi-supervised learning methods

	DrugBank	TTD	Average
Our method	39.78	12.26	26.02
w/o rule-filter	33.45 (−6.33)	6.82 (−5.44)	20.14 (−5.88)
w/o model-label	33.55 (−6.32)	11.09 (−1.17)	22.32 (−3.70)
DS w/interaction	34.04 (−5.74)	12.16 (−0.10)	23.10 (−2.92)

Note: We choose the Transformer + PubMedBERT-attn model and report the triplet *F*1 scores.

Table 6. Relation-extraction results on ChemProt

ChemProt	*F*1	Precision	Recall
SPINN model+pp	60.2	61.5	58.9
Tree-LSTM	58.5	67.0	51.9
Tree-LSTM+pp	61.7	65.7	58.1
Our method	61.4	67.4	56.4

Note: pp denotes additional pre-processing method used by the baseline methods.

5 Related work

Early research efforts on knowledge discovery are mainly extractive methods (Li and Ji, 2014; Miwa and Bansal, 2016; Zhong and Chen, 2020) and focus on discovering knowledge within a few sentences (Alt *et al.*, 2019; Miwa and Bansal, 2016; Zeng *et al.*, 2014; Zhang *et al.*, 2019). For document-level knowledge discovery, existing solutions are mainly graph-based methods that connect entities across sentences to handle knowledge expressed by multiple sentences (Christopoulou *et al.*, 2019; Nan *et al.*, 2020; Peng *et al.*, 2017; Quirk and Poon, 2017; Verga *et al.*, 2018). These methods often require detailed annotations, such as entity mentions and relations between mentions pairs, which are hard to obtain for DTI knowledge discovery. Recently, generative methods for knowledge discovery are explored (Ye *et al.*, 2021; Zeng *et al.*, 2020), which directly generate knowledge triplets from input sentences and only require end-to-end annotations. However, these works focus on sentence-level extraction. End-to-end generative extraction for document-level text like literature is still less investigated.

For discovering knowledge triplets from literature, early works attempt to extract simple relation between bio-entities. Li *et al.* (2016) extract relation between chemical substances and diseases, and Wu *et al.* (2019) focus relation between genes and diseases. The genes, diseases and chemical substances in those work are easier to recognize, and the extracted relationships are relatively simple (only one relation type) different from them, DTI covers much more diverse entity terms and more relations. Antunes and Matos (2019) and Döring *et al.* (2020) discover chemical proteins relation on document-level and sentence-level, respectively, which are related to DTI knowledge. Hong *et al.* (2020) explored to extract biomedical relation without any human annotation by using distant supervision technique. However, both of them mainly focus on relation extraction, where the entities are given in advance. By contrast, we aim to jointly discover the DTI triplets from the document. On the other hand, our datasets have more target and relational types and are much larger in volume.

6 Conclusions and future directions

In this work, we explore an end-to-end generative solution for extracting ⟨drug, target, interaction⟩ triplets from biomedical literature, which is one of the most important knowledge discovery tasks in the biomedical domain. Besides, we created KD-DTI, the largest dataset for this task, which is expected to boost and advance future research.

There are multiple directions to explore for this task:

1. *Accuracy improvement*: We have shown that the performance of several state-of-the-art models is still far from industry demand. Therefore, how to improve accuracy for the task is an important research problem. As shown in this article, designing better generative models and combing with pre-trained models properly are promising directions. How to effectively leverage unlabeled data (beyond pre-training) is also worthy of exploration.
2. *Dataset improvement*: We have created the largest dataset for the DTI discovery task. The dataset can be improved in terms of scale and quality. Furthermore, there are many other knowledge discovery tasks in the biomedical domain, which also need public datasets for algorithm evaluation.

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